TOPOLOGICAL INSULATORS — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 31, 2026

Diagram-based questions on Topological Insulators

Question 1

10.0 points

Examine the band structure diagram of a topological insulator in the diagram below, which shows band inversion between the bulk valence and conduction bands. Describe how this inversion results in a non-trivial topology.

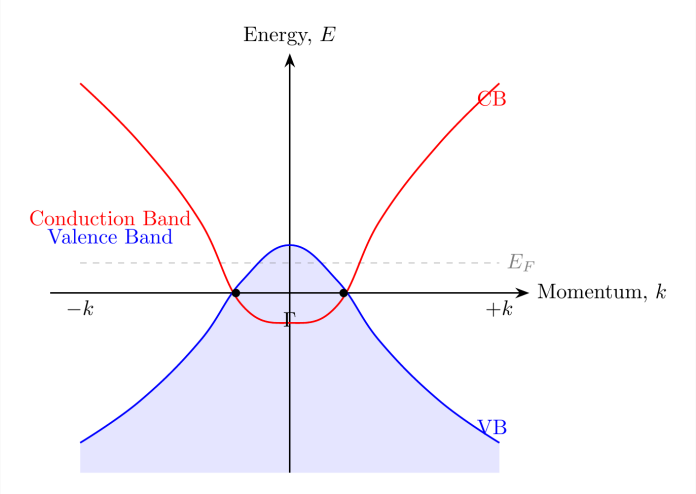

Fig. 1. A band structure diagram of a topological insulator showing band inversion. The diagram should include the bulk valence band and conduction band, with a clear indication of the band inversion. Label the energy axis and momentum axis. Highlight the point of inversion where the conduction band dips below the valence band, indicating a non-trivial topology. IMPORTANT correction needed: The diagram contains the label "Band Inversion" which. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

Question 2

10.0 points

Analyze the bulk-boundary correspondence as illustrated in the surface state diagram of a topological insulator in the diagram below. Explain how the surface states are protected by topology.

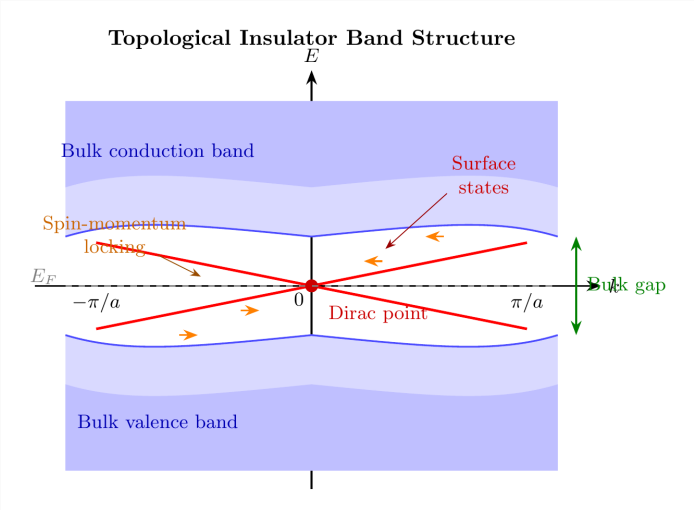

Fig. 2. A surface state diagram of a topological insulator showing the bulk-boundary correspondence. The diagram should include a representation of the bulk bands and the surface states. The bulk bands should be depicted as continuous regions in momentum space, while the surface states should appear as lines connecting the bulk bands across the band gap. Highlight the topological protection of the surface states by showing how they connect the valence and conduction bands, indicating the presence of a band gap in the bulk and gapless states on the surface.

Question 3 10.0 points

Analyze the edge state diagram below associated with the quantum spin Hall effect. Explain how these states allow for dissipationless edge transport.

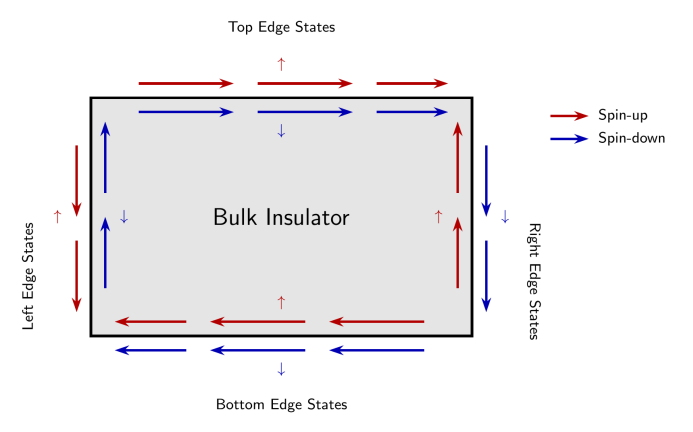

Fig. 3. A diagram illustrating the edge states in the quantum spin Hall effect. Show a topological insulator with edge states on the boundary. Indicate the direction of electron flow along the edges, with arrows. Represent spin-up and spin-down states with different colors or labels to show their separation. Include labels for the bulk insulator and the edge states to clarify the dissipationless transport mechanism. IMPORTANT correction needed: The diagram includes an annotation box that explicitly states the core. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram incorrectly. Regenerate using EXACTLY the values and sequences specified.

Question 4

10.0 points

Analyze the k-space structure of a 2D system with a Z2 topological invariant as depicted in the diagram below. Explain the significance of the Z2 invariant.

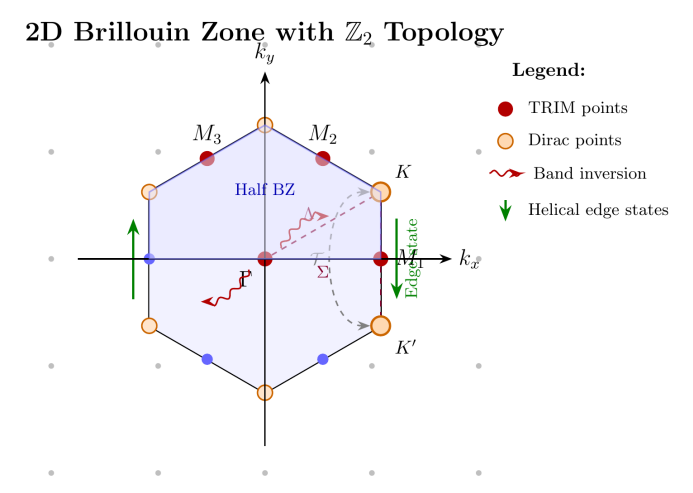

Fig. 4. A diagram of the k-space structure of a 2D system with a Z2 topological invariant. The diagram should show a 2D Brillouin zone with key features such as Dirac points or band inversions that contribute to the Z2 invariant. Highlight the topological boundary states and any symmetry lines or points that are relevant to the Z2 classification.

Question 5

10.0 points

Examine the Dirac cone dispersion relation depicted in the diagram below on the surface of a 3D topological insulator. Discuss how this structure is related to massless Dirac fermions.

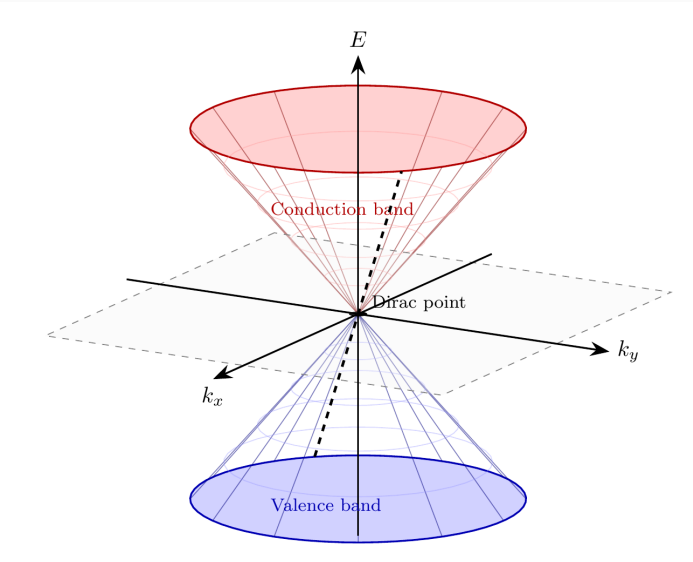

Fig. 5. A Dirac cone dispersion relation diagram showing the linear energy-momentum relationship on the surface of a 3D topological insulator. The diagram should include the Dirac point at the intersection of the cone, with energy (E) on the vertical axis and momentum (k) on the horizontal axes. The cone should be symmetric around the Dirac point, illustrating the linear dispersion characteristic of massless Dirac fermions.

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